

Rugved Zambare

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SUMMARY

Machine Learning and Data Science undergraduate skilled in Python, PyTorch, and HuggingFace Transformers, with hands-on experience fine-tuning LLMs, building computer vision and OCR pipelines, and deploying REST APIs on GCP Cloud Run, consistently improving model accuracy and latency.

EDUCATION

Pimpri Chinchwad College of Engineering, Pune	2023 to 2027
B.Tech, Computer Engineering Relevant Coursework: Machine Learning, Neural Networks, DSA, DBMS, Statistics	
Vidya Valley North Point Junior College, Pune	2021 to 2023
Kamalnayan Bajaj School, Pune	2009 to 2021

TECHNICAL SKILLS

Machine Learning & Deep Learning: PyTorch, CNNs, Transfer Learning, scikit-learn, ResNet18, Model Evaluation (F1, Precision/Recall, BLEU), AMP Mixed-Precision Training

Natural Language Processing & LLMs: HuggingFace Transformers, Qwen3, Fine-Tuning, RLHF, Supervised Fine-Tuning (SFT), Streaming Inference, LLM Evaluation

Computer Vision & OCR: OpenCV, MediaPipe, TrOCR, EasyOCR, ArUco Markers, HSV Segmentation, WebAR (model-viewer, AR Quick Look, Scene Viewer), glTF/USDZ

Data Science & Programming: Python, NumPy, Pandas, Statistics, JavaScript, C++, C

MLOps, Cloud & Backend: Git, GitHub, GCP (Cloud Run), FastAPI, Flask, REST APIs, SQLite, Experiment Tracking, CI-style Pipelines

Web & APIs: HTML5, CSS3, JavaScript (ES6+), Spotipy, Spotify Web API, OAuth 2.0 PKCE

PROJECTS

StackMind: Fine-Tuned LLM on Stack Overflow Dataset	Qwen3 HuggingFace FastAPI GCP Git
- Reduced hallucination rate ~35% vs. Qwen3 baseline by fine-tuning on 500K+ Q&A pairs with HuggingFace Transformers, RLHF-aligned formatting, and automated BLEU/F1 evaluation in a Git-versioned pipeline. - Cut model iteration time ~40% by replacing ad-hoc Colab notebooks with CI-style training loops, checkpoint versioning, and Precision/Recall tracking; deployed a streaming REST API (FastAPI) on GCP Cloud Run.	
AI Assignment Grader: GPU OCR and Auto-Scoring Engine	PyTorch TrOCR EasyOCR Flask SQLite
- Achieved 7 ms per-assignment inference on a 4 GB GTX 1650 using AMP mixed-precision and batched inference with a hybrid EasyOCR + fine-tuned TrOCR pipeline, tolerant of skewed phone photos via auto orientation detection. - Lifted OCR recall ~33% to 100% on noisy handwriting with a fragment-tolerant fuzzy matcher (difflib SequenceMatcher); ResNet18 score-bucket classifier at 90.8% accuracy, shipped end-to-end (OCR, scoring, SQLite, Flask, HTML/CSS/JS).	
NOVA: Gesture-Controlled Spotify Player	Python OpenCV MediaPipe Spotipy OAuth 2.0 PKCE
- Built a touchless Spotify controller using MediaPipe Hands (21 landmarks/hand) to classify real-time finger-state gestures via webcam for play, pause, skip, and volume, with zero accidental triggers via a hold-frame debouncer and cooldown logic. - Implemented full Spotify Connect integration via Spotipy and the Web Playback SDK, registering as a named Connect device using OAuth 2.0 PKCE (client-side only) across search, playlists, library, and queue.	
AR Food Menu ("The Golden Oak"): WebAR Restaurant Menu	FastAPI model-viewer OpenCV glTF/USDZ
- Built a no-install web AR menu rendering life-size 3D dishes onto the diner's real table via native AR (AR Quick Look on iOS, Scene Viewer on Android), backed by a FastAPI JSON API for per-dish nutrition, macros, and ingredient data. - Anchored AR content with OpenCV ArUco-marker detection and optimized .glb/.usdz models for mobile delivery; shipped end-to-end and deployed on Vercel and Netlify.	

EXPERIENCE

Google Developer Group on Campus (GDGOC), PCCOE Design & AI/ML Team Member	Sep 2024 to Apr 2025
- Upskilled 100+ students co-facilitating PyTorch, Git, and model evaluation workshops; 12+ student-led ML projects submitted at the semester showcase. - Delivered a production NFC/RFID attendance system with a REST API backend as a live club project, cutting manual attendance overhead ~80% across 3 departments.	
Spectrum 2024, PCCOE Annual Technical Fest Core Team Member	Feb 2024
Executed a 500+ attendee event with zero critical incidents, coordinating 8 volunteer teams and running campaigns that exceeded registration targets by 20%.	